Isogenous curves and ECDLP cost: vertical, horizontal and coordinate

effects **Question.** On the curves of one ℓ-isogeny volcano, with one ECDLP instance carried to all of them, does solving cost differ:

- **vertically**, between levels (crater against floor);
- **horizontally**, between curves of one level (one endomorphism ring, hence horizontally isogenous);
- **by coordinates**, between isomorphic models of one curve?

The methods are Pollard rho, BSGS and index calculus. The protocol was frozen before any data (`PROTOCOL.md`, v1). The design is in `PLAN.md`. The success criterion (CLAUDE.md) asks for the gap to rho, reported as the distance remaining.

Answer

Measured, PROTOCOL v1.1 (§ "Amendment"); 6 classes, 200 curves × 4 models: 43,200 measured runs (46,912 with the two control classes, each after one warm-up), every one verified; 34 sessions audited, 408/408 replays exact.

method	vertical: floor / crater, 90% CI	verdict (fixed wording)	horizontal: τ between curves of a level (null 95th %ile), p	coordinates: τ between isomorphic models (null 95th %ile), p
BSGS (`bsgs.negation`)	1.000 [1.000, 1.000]	identical cost on every curve and model for every paired target (exact, 12/12 targets, 800 workload curves)	0	0
rho (`rho.negation`)	1.022 [0.983, 1.062], p = 0.28	no level effect detected; **not equivalent within ±5%** (the interval reaches +6.2%)	1.7% (2.6%), Q p = 0.28, max|z| p = 0.73	2.6% (4.6%), p = 0.70
rho, A/A seeds	1.001 [0.965, 1.039], p = 0.94	the noise floor of the same design	0% (2.7%), p = 0.87	2.5% (4.6%), p = 0.75
IC relation yield (`ic.pipeline`, 32 smallest abscissae, MITM)	0.980 [0.957, 1.004], p = 0.20	**inconclusive**: the planted +5% floor control was not detected (§ Controls), though the interval lies inside ±5%	2.2% (1.4%), p = 0.0005	**4.7% (2.5%), p = 0.0005**
IC cost S (reported, not tested)	1.039 [1.008, 1.071], p = 0.061		2.9% (2.1%), p = 0.002	5.6% (3.7%), p = 0.0005

What it says:

1. **BSGS:** the cost is a function of the instance alone. Carrying the instance by isogeny, and changing coordinates, leaves it exactly unchanged on 800 workload curves (measured; it also follows from BSGS being generic).
2. **Rho:**
   - No vertical, horizontal or coordinate effect was detected. Every statistic sits inside its null band and the A/A arm reads the same.
   - The vertical bound is ±6.2%, not the ±5% the protocol aimed for. The design's predicted ±2% did not hold: with 1–11 crater curves per class, the floor/crater ratio is noisier than pooled solve counts suggest.
3. **Index calculus:**
   - **Horizontal (measured):** curves of one level differ in relation yield by $\tau \approx 2.2\%$. That is above the null, so it is significant.
   - **Coordinates (measured):** isomorphic models of one curve differ **more**, $\tau \approx 4.7\%$. By the protocol's rule this is a **coordinate effect**, not an isogeny effect. The factor base "the 32 points of smallest abscissa" is a different set of points in each model, and the yield follows that set, not the curve.
   - **Vertical:** the pooled test cannot see a 5% effect (failed analysis control), so it is inconclusive. Exploratory per-class contrasts (§ Exploratory) suggest an $\ell = 2$-only effect of about −6.5%.
4. **Gap to rho (distance remaining):** on every one of the 800 workload curves, rho is below IC at every weight of the §5a band (0, 0.1 and 1 GAE per unpriced native operation; `band/`).
   - Per curve, IC costs $S \approx 27$–96 at $w = 0$ against rho's 0.8–1.6.
   - So IC is about 30–60× above rho at 24 bits.
   - No isogenous curve or model closes any part of that gap beyond the few-percent effects above.

Scope: 24-bit p, depth-3 volcanoes, $\ell \in \{2, 3\}$, the three methods at these parameters. No security statement. Nothing above 24 bits, and no other factor base, is covered.

Controls

control	measured	predicted	verdict

planted IC: factor base 33 vs 32 points, yield ratio matched over 3,200 workloads	1.0613 [1.0583, 1.0644]	$(33/32)^2 \approx 1.063$	detected; instrument sees +6%
j = 0 crater: `rho.cubic` / `rho-neg-crater`, 64 workloads	0.697 [0.589, 0.849]	$\sqrt{(1/3)} \approx 0.58$	detected (excludes 1)
j = 1728 crater: `rho.quartic` / `rho-neg-crater`	0.707 [0.604, 0.815]	$\sqrt{(1/2)} \approx 0.71$	detected (reported only)
BSGS exactness, all classes	1 distinct cost per paired target	1	exact
A/A rho (independent measurement seed), matched pair	0.9915 [0.9795, 1.0032]	1	within noise
analysis: +5% planted on every floor curve, rho	V 1.022 ⟶ 1.073, p = 0.0006	detect	**detected**
analysis: +5% planted on every floor curve, IC yield	V 0.980 ⟶ 1.029, p = 0.073	detect at 0.0125	**not detected** ⟶ IC vertical inconclusive
analysis: +10% planted on one curve per class, rho	max|z| p 0.73 ⟶ 0.73	detect	**not detected** ⟶ single-curve rho outliers below ±15% are not excluded
analysis: +10% planted on one curve per class, IC yield	max|z| p 0.0105 ⟶ 0.0005	detect	detected

The planted-control runs are in `analysis-controls/`, for both the v1 and v1.1 statistics.

Amendment v1.1 (after seeing data, stated as such)

BSGS cost is identical on every curve, but the frozen v1 statistic gave its floor/crater ratio a bootstrap interval of [0.88, 1.22].

- **Why:** v1 pools all classes into one floor mean and one crater mean. The classes contribute different shares of floor and crater curves. So when a bootstrap replicate redraws a class's paired targets and shifts that class's mean, the ratio moves even though no curve differs from another.
- **Fix:** v1.1 forms the ratio inside each class and combines the six ratios as an equal-weight mean of logs (`volcano-stats --vertical stratified`). The permutation tests are valid under either statistic.

Both versions are reported:

arm · metric	v1 pooled: V, 90% CI, p	v1.1 stratified: V, 90% CI, p
BSGS S	1.000 [0.881, 1.224], 1.0	1.000 [1.000, 1.000], 1.0
rho S	1.024 [0.988, 1.062], 0.071	1.022 [0.983, 1.062], 0.275
rho S, A/A	0.978 [0.947, 1.010], 0.069	1.001 [0.965, 1.039], 0.938
IC yield	**0.972 [0.951, 0.993], 0.0012**	0.980 [0.957, 1.004], 0.202
IC S	1.029 [0.998, 1.060], 0.024	1.039 [1.008, 1.071], 0.061

Under v1 the IC-yield vertical test is nominally significant (p = 0.0012 < 0.0125). The same planted analysis control fails under v1 too (+5% detected only at p = 0.081). Under v1.1, which removes the class-mix artefact the BSGS check exposed, it is not significant. By the frozen v1 rule the v1 result would read "level changes IC yield by −2.8%". This report does not claim that, because the statistic behind it is shown wrong by an exact control. The v1 numbers stay here and in `analysis.json`.

Exploratory (not tested; six unadjusted contrasts)

Per-class floor/crater, v1.1 statistic (`analysis-exploratory/`):

class	E(F_p) 2-part: floor vs above	rho S	IC relation yield
`l2-ramified`	cyclic vs Z/2 × Z/2	0.996 [0.889, 1.124]	**0.930 [0.863, 0.994]**, p = 0.033
`l2-split`	cyclic vs Z/2 × Z/2	1.043 [0.960, 1.139]	**0.937 [0.888, 0.987]**, p = 0.001
`l2-inert` (1 crater curve)	cyclic vs Z/2 × Z/2	1.074 [0.931, 1.225]	0.988 [0.906, 1.086]
`l3-ramified`	all cyclic	1.012 [0.897, 1.130]	0.964 [0.913, 1.018]
`l3-split`	all cyclic	1.019 [0.940, 1.110]	1.003 [0.958, 1.052]
`l3-inert`	all cyclic	0.991 [0.860, 1.140]	1.066 [0.983, 1.165]

Hypothesis for PROTOCOL v2 (not a finding).

- **Pattern:** in $\ell = 2$ volcanoes, floor curves (cyclic 2-part, one rational 2-torsion point) give about 6.5% fewer IC relations per trial than crater curves (full rational 2-torsion). $\ell = 3$ volcanoes, whose groups are cyclic at every level, show nothing.
- **Mechanism to test:** the small-abscissa factor base is drawn from the whole group, and how its points split across the 2-power torsion cosets differs with the group structure.
- **What would confirm it:** fresh $\ell = 2$ classes with several crater curves each, a pre-registered one-sided test, and the IC cost S on the same curves.
- **What would make it an improvement:** a confirmed yield gain also has to show in S and survive the §5a band before it counts as an index-calculus improvement (CLAUDE.md success criterion).

Population and instrument

Six ordinary classes plus two positive-control classes, all 24-bit, depth 3. The table (p, root model, curves per level, group structures) is in `PROTOCOL.md` §2.

- **Curves and transport:** `volcano-rho --ecbench` exported every curve of each volcano, up to 12 per level, with every crater curve kept. Each curve came in 4 isomorphic models, with the isogeny-transported generator and `planted_shared` targets, so every curve and model solved the same k.
- **Runs:** `ecbench` (L0, counted units only); each session's command and spec are in `classes/` and `sessions/`.
- **Analysis:** `volcano-stats` (native), with bootstrap 2,000 and permutations 10,000 (2,000 for the heterogeneity nulls). Files: `analysis-v1.1.json` (headline), `analysis.json` (v1 as frozen) and `analysis-j0.json` / `analysis-j1728.json` (controls).
- **Figures:** `plot.py` ⟶ `figures/`: one volcano heat map per class with rho cost and IC yield panels, `vertical_forest` and `heterogeneity`.

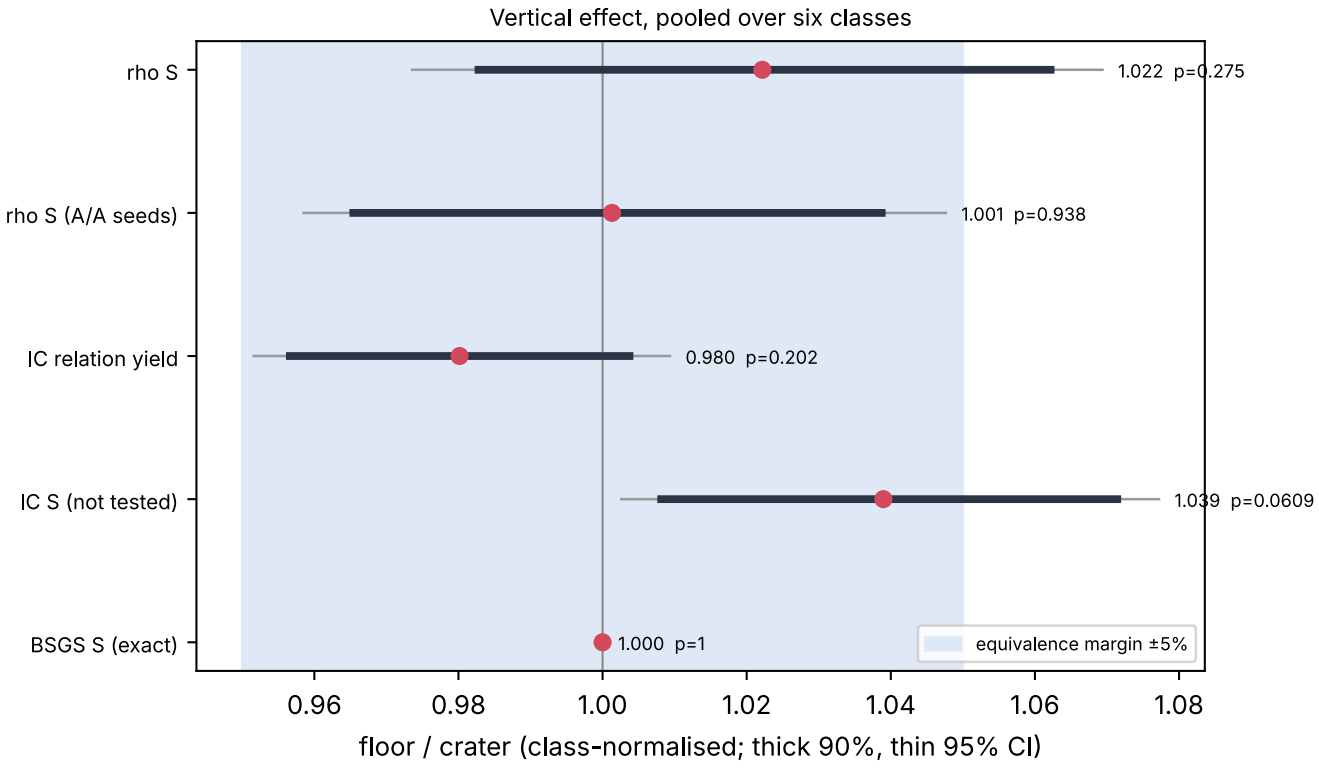


Figure 1: vertical

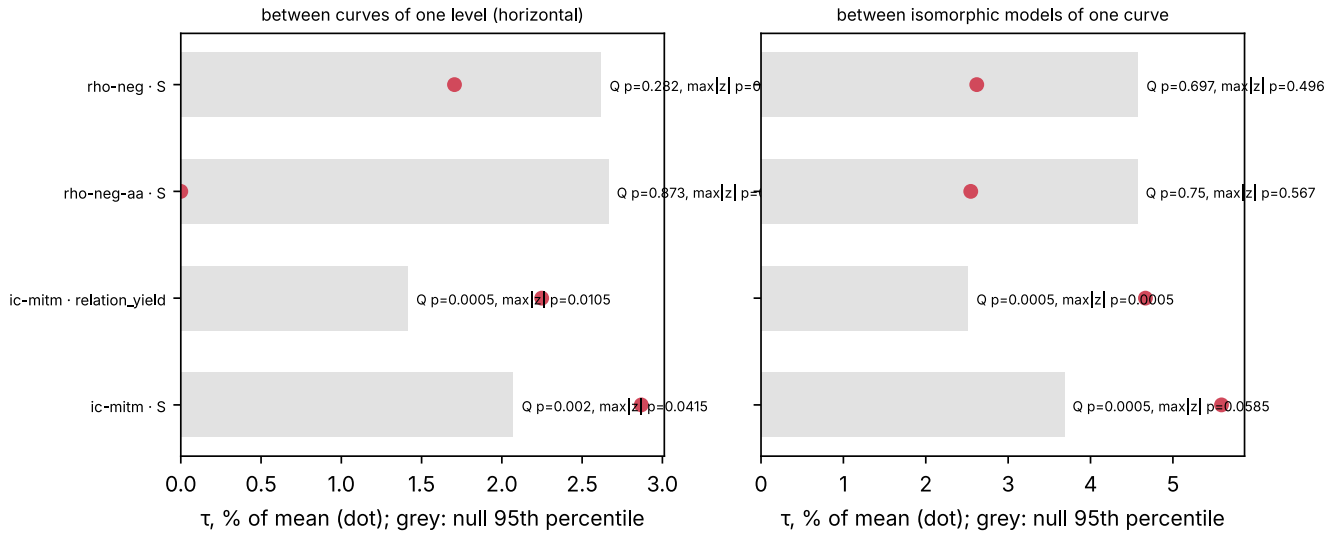


Figure 2: heterogeneity

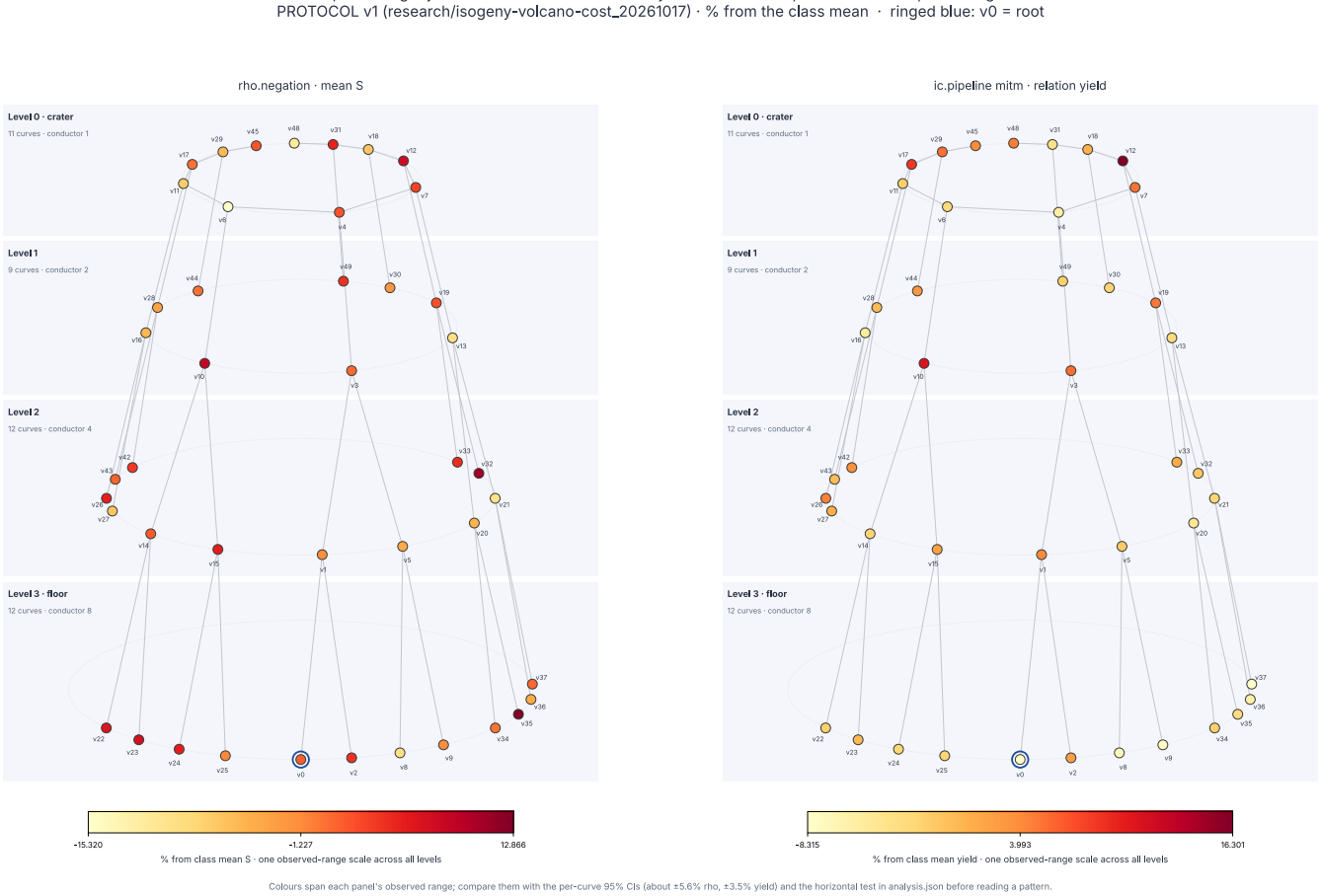


Figure 3: l2-split

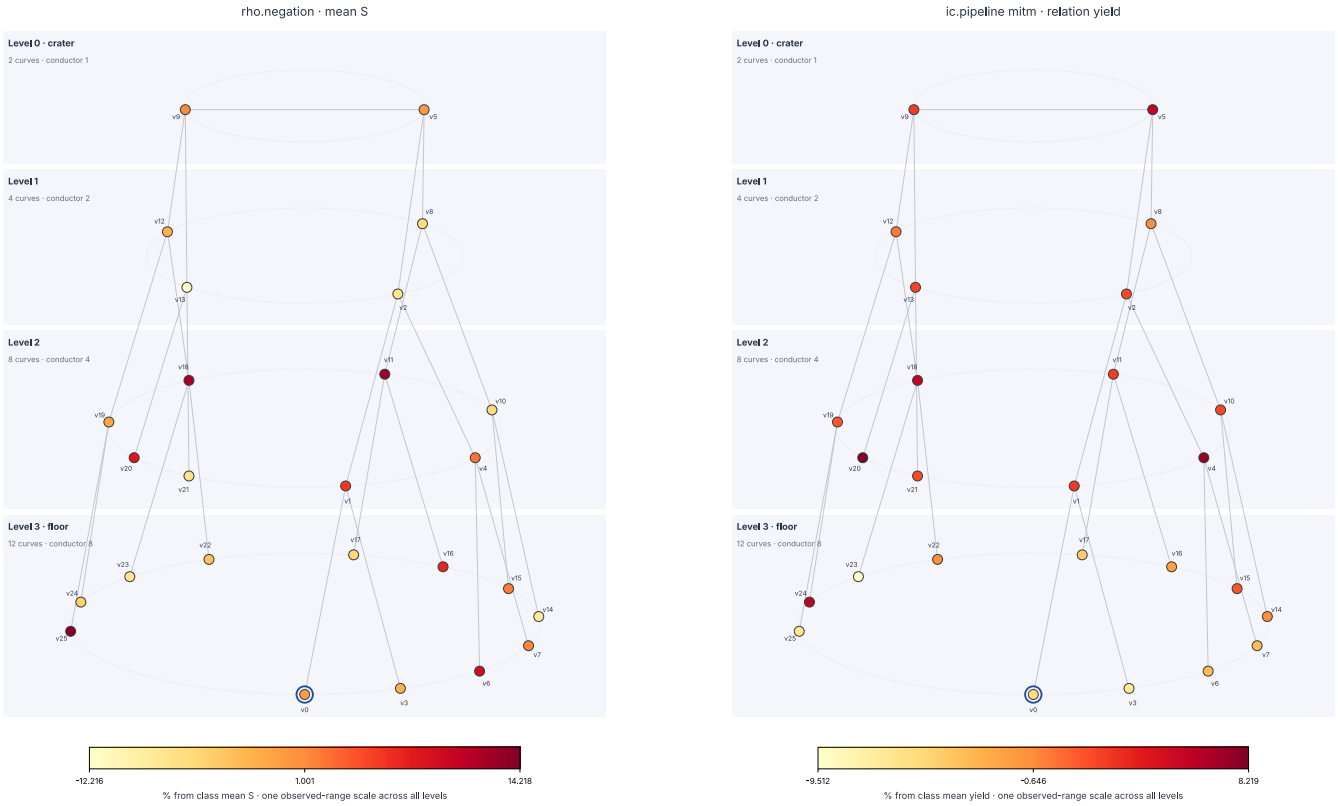


Figure 4: l2-ramified

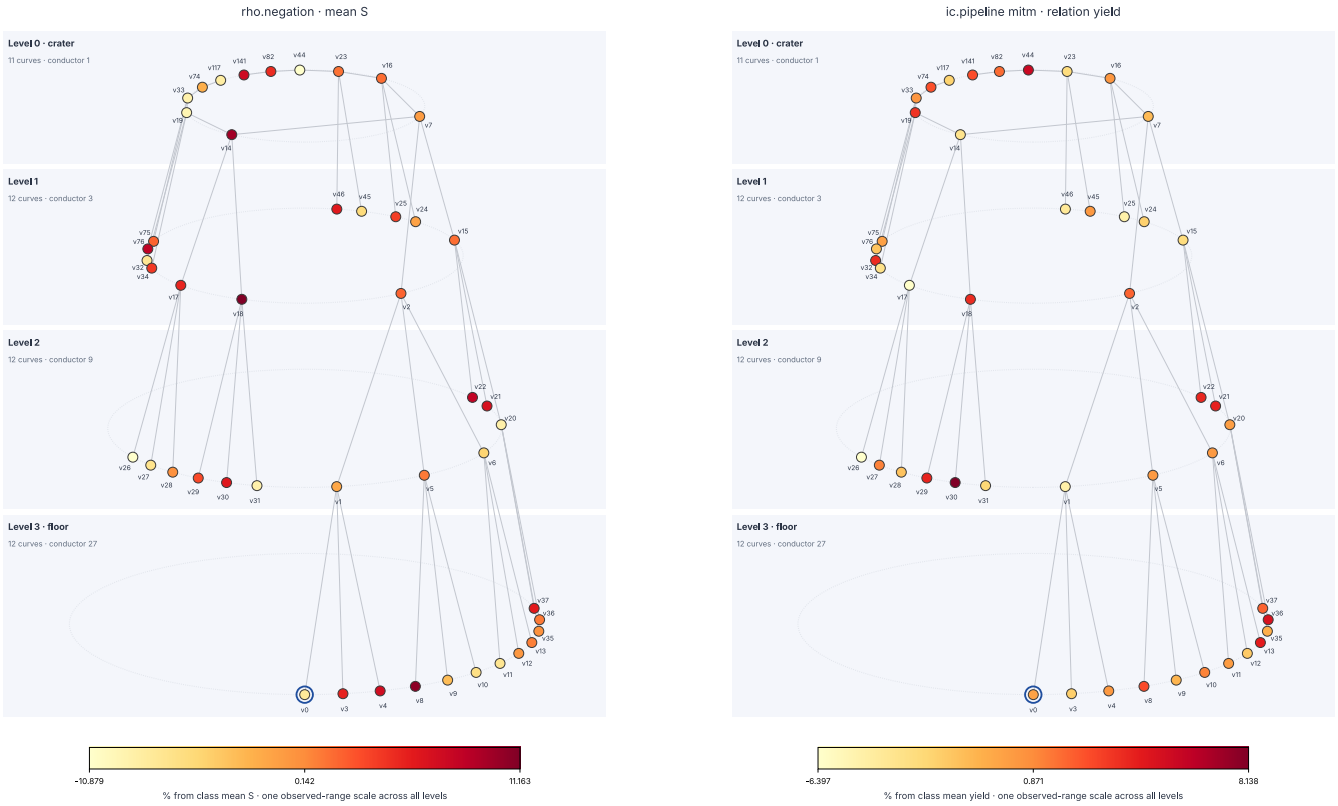


Figure 5: l3-split

Evidence and limits

- **Sessions:** 34 complete sessions, 46,912 measured runs (each after one warm-up), every run `verified`. `ecbench verify --replay 12` passed on every session: 408/408 replays identical. Audits are in `audits/`.
  - `l3-split/spec` was interrupted at 7,784/12,032 runs by the host's 2-hour background limit, not by its result. It stays as evidence; the analysis uses its complete re-run `l3-split/spec-r2`.
  - The two BSGS sessions the limit prevented from starting ran afterwards.
- **Storage:** records and plans are gzipped (`*.jsonl.gz`, `plan.json.gz`; `MANIFEST.sha256` holds the uncompressed hashes). Decompress before `ecbench verify`.
- **Isolation:** the host is a shared 4-vCPU VM at L0. No wall time is cited. Counted units are host-independent. Independent-host replay is not done, so this is not a claim under the IC claim rules, and it makes no IC-beats-rho claim.
- **Power:**
  - The rho vertical bound is ±6.2%.
  - IC vertical is inconclusive.
  - Rho single-curve outliers below about ±15% are not excluded: per-curve rho has 64 runs, about ±11% at 95%.
  - More crater curves (more classes, more models) is the lever for all three.
- **Not run (protocol v2):** the 20- and 28-bit size arms, and rho at 32 and 40 bits (PLAN §4).
- **Conductor:** not installed in this container, so no task was reserved.